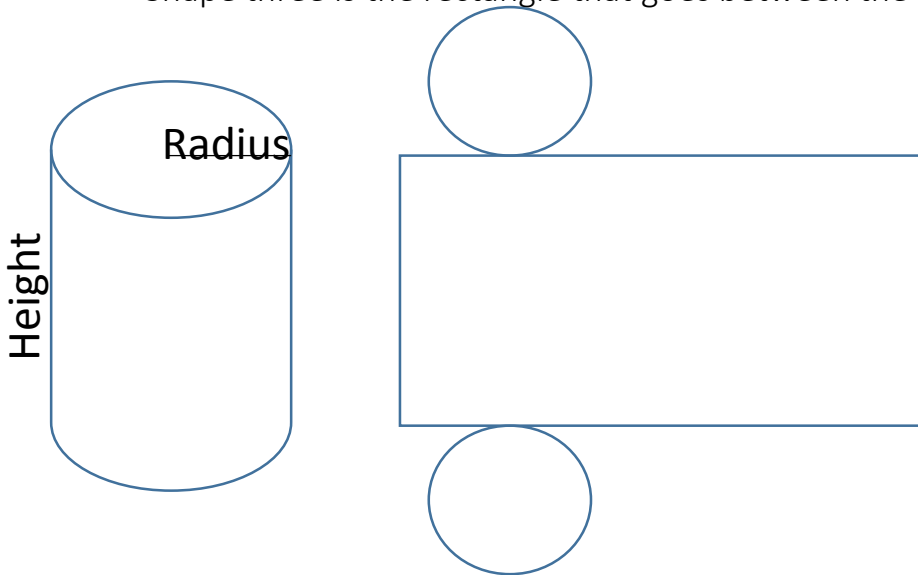


## How do we find the Surface area of a cylinder?

The surface area of cylinder is made of three shapes.

Shape 1 and 2 are the circles on the top and bottom of the cylinder.

Shape three is the rectangle that goes between the cylinder



The area of a Cylinder is worked out by adding the area of the 2 circles and the area of the rectangle.

The area of a circle is found using the formula for the area of circle.

$$\text{Area of a circle} = \pi \times \text{Radius}^2$$

The area of the rectangle is found using the formula;

$$\text{Area} = \text{length} \times \text{height}.$$

In a cylinder the length is equal to the circumference of a circle. The circumference is calculated using the formula;

$$\text{Circumference} = 2 \times \pi \times \text{Radius}$$

Total surface area of a cylinder = 2 x area of the circle plus the area of the rectangle

$$\text{TSA} = 2 \times \pi \times \text{Radius}^2 + 2 \times \pi \times \text{Radius} \times \text{Height}$$

Find the Total surface area for each of the following cylinders

